STAGE 2 – PRACTICAL ENGINEERING CHALLENGE

TEAM ID:	TEAM NAME:
LRC-26-0344	It is Bot it is

Problem Statement: Industrial Manufacturing Process Automations

Industrial manufacturing depends on highly controlled liquid-handling systems to deliver process fluids reliably from a source or bulk supply to downstream process equipment. Depending on the application, these fluids may include **ultrapure fluids, process chemicals, solvents, developers, or CMP-related fluids**. Industry standards explicitly address liquid-chemical distribution, high-purity piping, bulk chemical distribution, system qualification, and UPW system operation and monitoring.

In such systems, **flow is a fundamental process variable**. An incorrect flow rate can indicate or cause a restriction, incorrect valve state, inadequate pumping performance, abnormal process demand, or another hydraulic deviation. The consequence may be unstable process conditions, incorrect chemical delivery, equipment malfunction, or loss of process performance.

This challenge therefore focuses on a fundamental industrial facilities and process-engineering task: **To establish a defined process-fluid flow, measure it reliably, detect a deviation, diagnose its cause, and restore the process to the required operating condition.**

Engineering Challenge

As you are part of a multidisciplinary engineering team responsible for designing, building and validating a small-scale automated liquid-flow system represented of industrial process handling system.

CASE 1 – CONTROL + MEASUREMNT

Problem: Develop a system that manages the changing level of fluid in the sump automatically (when there is a sudden external fluid charging to the sump) and while accurately measuring and totalizing the fluid discharged from the sump using the flow transmitter.

CASE 2 – FLUSHING +SHUTDOWN

Problem: Develop an alternative operating configuration that allows flush water to pass through the system while all three pumps operate simultaneously, and then safely transition the system to an OFF condition when the flush-water supply has been exhausted.

Given Process Parameters

Pipe diameter =	**8mm**
Fluid density (ρ) =	**1000kg/m^3**
Dynamic Viscosity (μ) =	**1x10^{-3}Pa.s**
Required Flow range =	**1-6 L/min**
Nominal target flow for the live challenge =	**3L/min**

Available Hardware

Mechanical Hardware	Electronic Hardware
1. Water Tank – 3 Quantity	1. Custom PCB
2. Pumps – 3 Quantity	2. ESP32 NodeMCU module
3. FT-01 Flow Transmitter – 01 Quantity	3. Pump Driver-1
4. Ball valve – 3 Quantity	4. Pump Driver-2
5. ON/OFF/XV Valve – 1 Quantity	5. Pump Driver-3
6. 10-mm OD tubing/blue – 9 meters/1 meter	6. 12 V DC power input (**SMPS**)
7. Microcontroller/PCB/ data-acquisition software	7. 12 VDC Solenoid-valve interface
8. Required electrical/ electronic accessories	8. Flow-sensor interface
	9. Float-switch interface
	10. ON/OFF-switch interface
	11. Pump-driver PWM/enable interfaces
	12. On/Off Switch

The team must determine the appropriate arrangement, connections and measurement method.

- Assemble and commission the liquid-flow system using the components provided.
- Measure and characterize the flow using FT-01
- Interface FT-01 with the provided **data-acquisition system**
- Experimentally determine the relationship between sensor output and actual flow.
- Characterize the effect of valve position on flow rate. Record and analyse flow vs time data
- Maintain target flow of 3L/min
- Use graphs and measured data to justify engineering decisions
- Design an appropriate hydraulic arrangement and P&ID

TEAM INSTRUCTIONS

- As a team go through the list provided to read about each component that has been Listed.
- The PCB is a 12 V DC and the SMPS connections should be made carefully.

TASK 1: HYDRAULICS SYSTEM DESIGN

Q1.1 Design and build the hydraulic path with the flow transmitter on the recirculation line. Submit/sketch a labelled P&ID below by showing the pumps, Needle valves, flow transmitter, flow direction, Tank, sump, tube diameter and measurement points.

CASE 1 – Process Fluid Transfer & Automatic Level Protection

Objective: Set-up and operate a closed-loop process- fluid handling system using the available tank, pump, valves flow transmitter, sump, float sensor and control PCB. The System must continuously transfer process fluid recirculation system by connecting the 10L fluid tank with a On/Off valve to a pump-01 and the pump-01 goes to the inlet of sump 4.2L tank. The Sump 4.2L outlet line is connected to another pump-02 followed by a flow transmitter that measures the total flow back to the 10L fluid tank. The Pump-01 runs on 50% duty cycle and Pump-02 runs on 100% duty cycle. **Calibrate the flow transmitter** and demonstrate correct fluid circulation, if there is sudden introduction of external fluid into the Sump tank then Pump-01 should stop while pump – 02 should still runs till fluid level reduces in the sump. After the fluid reduction in the SUMP 4.2L tank the Pump-01 should turn on and the process must continue. **Also, demonstrate Flow vs Time trends in mL/s, total transferred Liters back to the 10L tank** (share Screenshots flow in L/min). **Share the code**. The case-1 can be turned on using the available on/off Switch, to start the system.

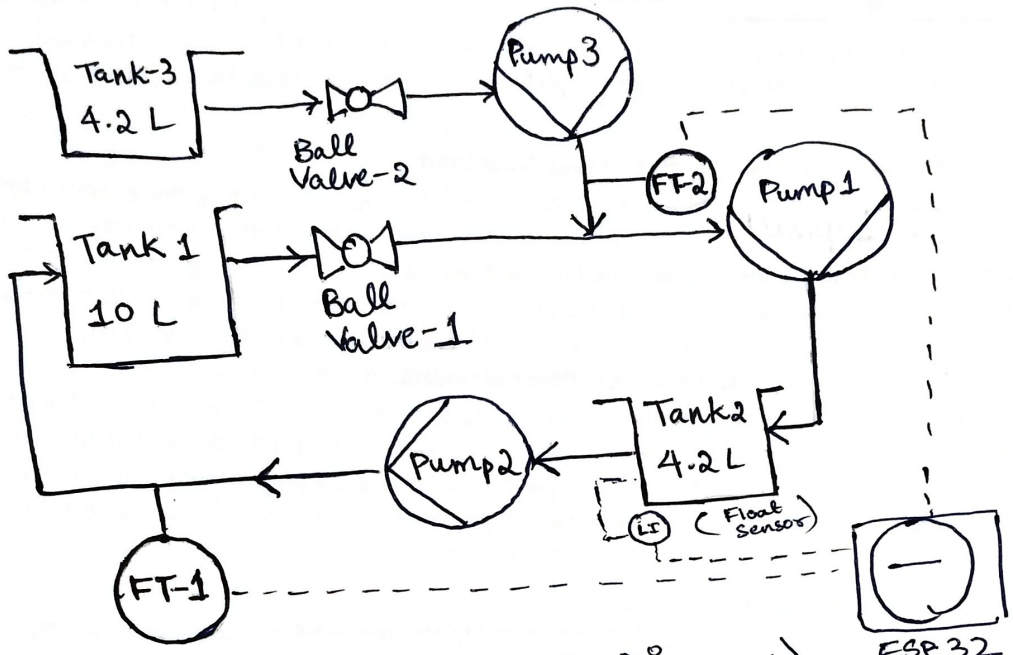
CASE 2 – Flush/Line Conditioning Sequence

Objective: After successfully completing case 1, initiate the flush water in the sequence using the **equal Tee.** The On/Off valve must be switched off in order to isolate the 10L process fluid tank. The flush Fluid tank (4.2L) is used as a temporary fluid source to flush out the process fluid or any debris from the system. Before the start of this process make sure you open the manual ball valve on the fluid flush line before Pump-03 starts. The entire process of case -1 and case - 2 should run in a single go without manual intervention except opening of ball valve before the Pump-03 starts. **The System must stop as soon as the tank flush fluid 4.2L tank finishes.** The system must be automated without human interventions (only ball valve can be opened manually before the start of flush line sequence)

SUBMIT:

A. Submit: 10-15 min Video and Photograph of the completed setup, with identification of each component and explain the direction of flow in the video verbally. Demonstrate both Case - 1 and Case – 2.

B. Final P&ID Block diagram



(Piping & Instrumentation Diagram)

3 Modes in the system:

Mode 1: (Case 1) Pump 1 ON
Pump 2 ON
Pump 3 OFF

Mode 2: (Stand By) All Pump OFF

Mode 3: (Case 2) Pump 1 ON
Pump 2 ON
Pump 3 ON

In mode 1, when the float sensor detects full tank, Pump 1 is turned OFF.

In mode 3, when the flow transmitter (FT-2) detects very low pulse rate (i.e empty tank), the system automatically turns off by going into Mode 2.

Q1.2 After setting-up the system, now operate it at approximately at 1L/min, 3L/min & 6L/min. Compare the theoretical velocity with the actual operating behaviour.

WORKING / COMPARISON

~~The maximum observed flow rate is 3L/min. Hence we~~

For this experiment we varied the pump duty cycle ~~at arbitrary~~ at 33%, 66% and 100% to get a theoretical flow rate of 1, 2, and 3 L/min respectively

	Measured (flowmeter)	Measured (jug) (experimental)	Expected
33%	15.3	13.33	16.6
66%	34.9	32.25	33.3
100%	46.8	47.84	50

All values in mL/s

We can observe that the actual (jug) flow is much less than expected flow. ~~For a % of the~~ The percentage drop in flow is also increasing as flow rate decreases. This indicates that resistances act much more harshly at lower velocity

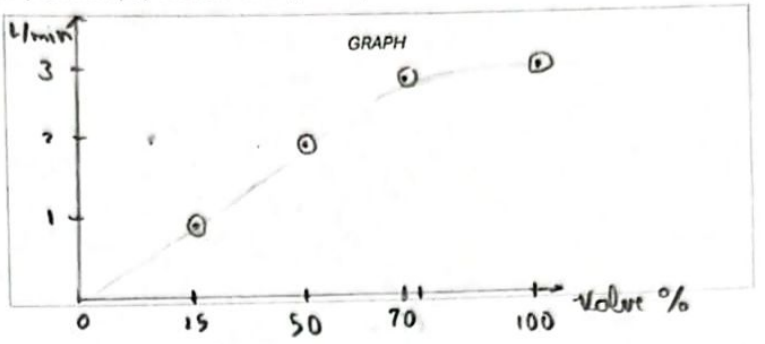
The flowmeter also gives ~~worse~~ more flow than actual, ~~and~~ except in the case of 100%. This is likely due to ~~to~~ effects of turbulence ~~of~~ at higher flow rates.

Note :- The maximum observed flow rate is 3L/min, hence we are using flow values of ~~1,~~ 1, 2, 3 L/min

TASK 2: FLOW RESPONSE TO VALVE POSITION

Q 2.1 Set the ball valve to approximately 25%,50%,75% & 100%. At each position, allow the flow to stabilize and record the flow rate. Create the following graph of valve position (%) vs Flow Rate (L/min)

VALVE POSITION	FLOW RATE
25%	0.91
50%	1.95
70%	2.78
100%	2.9



Q2.2 Determine the experimental calibration equation: $Q = af + b$, where Q = flow rate in L/min, f = pulse frequency in Hz, a = experimental calibration coefficient and b = offset.

Report:

R^2 = 0.9934

a = 0.02352

b = −0.1028

Maximum calibration error = −0.108

Average calibration error = 0.0498 [Note :- Average absolute error is 0.0498, average error is 0]

Q2.3 FT-01 produces pluses corresponding to water flow. Run the system at least for 5 different flow conditions within the available operating range (Hint: use formulas f=N/t and Q=V/t). For each condition, record the following:

Trial	Pulses	Time(s)	Frequency (Hz)	Collected Volume (L)	Actual Flow (L/min)
1	2616	20.7	126	1	2.9
2	2649	21.6	122	1	2.78
3	1359	15.4	88	0.5	1.95
4	1456	19.6	74	0.5	1.53
5	1317	32.9	40	0.5	0.91

CALCULATIONS / OBSERVATIONS

Least squares fit :- Treating x (%) as y (L/min)

x	y	x^2	xy	y^2
40	0.91	1600	36.4	0.8281
74	1.53	5476	113.22	2.3409
88	1.95	7744	171.60	3.8025
122	2.78	14884	339.16	7.7284
126	2.90	15876	365.40	8.41
Σ 450	10.07	45580	1025.78	23.1099

$$m = \frac{(n\Sigma xy - \Sigma x \cdot \Sigma y)}{(n\Sigma x^2 - (\Sigma x)^2)} = 0.02352$$

$$c = \Sigma y - m\Sigma x = -0.1028$$

x	y_{pred}	y_{act}	Error
40	0.838	0.91	0.072
74	1.638	1.53	-0.108
88	1.967	1.95	-0.017
122	2.767	2.78	0.013
126	2.861	2.9	0.039

$$R = \frac{n\cdot\Sigma xy - \Sigma x \Sigma y}{\sqrt{(n\Sigma x^2 - (\Sigma x)^2)(n\Sigma y^2 - (\Sigma y)^2)}} = 0.9967$$

$R^2 = 0.9934$

Mean absolute error $= \frac{\Sigma|e|}{n} = \frac{0.2491}{5} = 0.0498$

2.1 :- The valve actuation vs. flow is largely linear, until about 70%, at which point the flow saturates

2.3 :- The 95% confidence interval for the intercept is $[-0.96, 0.61]$, hence it is valid to use 0 in the code. This intercept is likely because of a nonlinearity in the sensor near the 0 flow point.

TASK 3: HYDRAULIC SYSTEM THEORY AND FAULT ANALYSIS

Q3.1 The effective liquid volume inside a section of tubing is 0.5L, while the operating flow rate is 2.5L/min. The hydraulic residence time is given by $T_r = Q/V$. Calculate the residence time in seconds, also answer the following questions:

1. What does residence time physically represent in this system?
2. What happens to residence time if the tubing volume is doubled while the flow rate remains constant?
3. What happens to residence time if the flow rate is doubled while the tubing volume remains constant?
4. Why is residence time important when interpreting a sudden change in measured flow?

ANSWER

* Correct formula for T_r would be V/Q. Using this for further calculations.

Given: $V = 0.5\,L$; $Q = 2.5\,L/min$

$$\Rightarrow T_r \text{ (in s)} = \frac{0.5}{2.5} \times 60 = \boxed{12 \text{ secs}}$$

1. Physical meaning: The average time a fluid element takes to travel through the section of tubing. It can also be interpreted as the time taken to replace the whole volume once.

2. $T_r = \frac{V}{Q}$ if $V' \Rightarrow 2V$ while Q remains constant

$$\Rightarrow T_r' = \frac{2V}{Q} = 2T_r$$ i.e. residence time doubles.

3. $T_r = \frac{V}{Q}$ if $Q' = 2Q$ & $V' = V$

$$\Rightarrow T_r' = \frac{V}{2Q} = \frac{T_r}{2}$$ i.e residence time halves.

4. Residence Time is important for interpreting sudden flow change because:
   (1) A pump tripping or valve restriction upstream will take about one T_r time to show up in the reading of flow transmitter. instead of showing up instantly.
   (2) To find the cause of this change in flow rate too we need to account for this residence time/delay or we might end up blaming the wrong component.

Q3.2 The following observation are obtained from the same system:

CONDITION	PUMP	VALVE	FT-01 PULSE SIGNAL	MEASURED FLOW
A	ON	Open	Normal	Normal
B	ON	Partially closed	Reduced	Reduced
C	ON	Open	Zero	Zero
D	OFF	Open	Zero	Zero
E	ON	Open	Intermittent	Fluctuating

For each of the condition, identify the most likely physical or instrumentation condition. Then answer: Which condition is most difficult to distinguish using FT-01 alone, and what additional information would you require to distinguish the fault? (Hint: Teams must consider possibilities such as: Valve restriction, pump failure, hydraulic blockage, sensor/signal failure, unstable flow etc)

Condition: Likely Causes

A — Normal System Operation
The pump is operating correctly, the valve is allowing sufficient fluid to pass & FT-01 generates normal pulse.

B — Valve Restriction
Valve is partially closed, FT-01 shows correct flow.

C — Complete Blockage / Pump Failure / FT-01 failure

D — Normal Stand By / Shutdown condition.
Pump is OFF, so zero flow is expected

E — Unstable Flow could be caused by: air in the system, partially blocked lines, wiring fault,

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'C' is the most difficult to diagonose using FT-01 alone because zero pulse signal & zero measured flow donot help estaldish the cause of the problem. Additional information required would be:

- Pump motor current : High current could mean blocked pump.
  Very Low current would mean pump is dead
- Tank & Sump Levels : If levels are changing ⇒ FT-01 has failed

HARDWARE LIST

Mechanical Hardware

Component	Description	Main Use
Ball Valve (BV-01)	A manually operated valve with a spherical shut-off mechanism.	Used to **manually isolate or permit fluid flow** in the pipeline. It is useful for maintenance, testing, and flow isolation.
10-Litre Fluid Tank (Fluid Tank-01)	Transparent storage tank with approximately 10 L capacity.	Used as the **main fluid reservoir** for storing process fluid. The transparent body also allows the fluid level and condition to be visually monitored.
5-Litre Sump	Small transparent reservoir labelled **SUMP**, approximately 4.5 L.	Used as a **collection/storage vessel** for fluid returning to or entering the sump section of the system.
4.2-Litre Flush Fluid Tank / Flush Line	The transparent vessel labelled **FLUSH FLUID** is approximately 4.5 L.	Provides **flush fluid** for cleaning or flushing the main pipe circuit when required.
ON/OFF Valve (OV-01)	Electrically operated valve/solenoid-type valve shown in the setup.	Used for **automatic control of fluid flow**, allowing the controller to turn a particular flow path on or off.
Pumps -Pump-1, Pump-2, Pump-3	Motor-driven fluid pumps. Three pumps are shown in the setup.	Used to **circulate and transfer fluid** between different sections of the system. Each pump can be independently controlled.
8 mm Process Pipe	Flexible blue tubing with approximately 8 mm process-line size.	Used as the **main fluid-flow path** connecting tanks, valves, pumps, tees, and other components.
10 mm × 8 mm Reducers	Reducer fittings that transition between a 10 mm connection and an 8 mm pipe.	Used to **adapt the pump connection size to the 8 mm process pipe**. **Mainly used for Pumps**
8 mm Connectors	Push-fit/barbed-type fittings compatible with the 8 mm tubing.	Used to **join pipes to components** and to make detachable connections between sections of tubing.
8 mm Equal Tee	T-shaped connector having three equal 8 mm ports.	Used to **split one 8 mm fluid line into two branches or combine two branches into one line**.

Electronic Hardware

Electronic Component	Identification / Function	Use
PCB -Printed Circuit Board	Custom control board	Provides the electrical interconnection between the **12 V power supply, ESP32 controller, sensors, valves, indicators, and pump-driver interfaces**.
ESP32 / NodeMCU ESP32S module (U7)	Main microcontroller	Acts as the **central controller**. It provides GPIO/PWM signals for controlling outputs and receiving signals from sensors/switches.

Electronic Component	Identification / Function	Use
12 V DC Input Connector	Marked **12V INPUT**	Connection point for the external 12 V DC power supply.
12 V Output Connectors	P1, P2, P3 and additional 12 V output terminals	Provide 12 V power connections for external loads such as the pump-driver/motor circuits.
Motor Driver Interfaces -U2, U3, U4	Three multi-pin driver connectors shown on the PCB	Provide dedicated interfaces for **Pump-1, Pump-2 and Pump-3 driver/control connections**.
Solenoid Valve Connector	Labelled **SOLENOID_VALVE -S/V/G**	Interface for controlling/connecting the solenoid or ON/OFF valve.
Flow Sensor Connector	Labelled **FLOW_SENSOR - S/V/G**	Interface for the water-flow sensor; supplies power and receives the sensor signal. **THE FLOW SENSOR SHOULD BE CALIBRATED**
Float Switch Connector	Labelled **FLOAT_SWITCH -FSW/GND**	Interface for a float/level switch used to detect a water-level condition.
ON/OFF Switch Connector	Labelled **ON/OFF_SWITCH -SW/GND**	Interface for an external manual ON/OFF switch.
12V DC SMPS	**12V DC SMPS**	For Power Supply to the board

Code For ESP32:

```
// Button cycles:  0 = motors 1+3 ON (case 1)
//                 1 = all OFF
//                 2 = motors 2+3 ON (flush)
//                 3 = same as state 1

// Float conducting = motor 1 off, and stays off for HOLD_MS after it opens.
// Flush: after GRACE_MS in mode 2, if the flush-tank flow stays below
// FLUSH_MIN_LMIN over a 500 ms window, the code goes to mode 3.

// ============== SETTINGS ==============
int POWER1 = 50;                 // pump 1 motor from main tank
int POWER2 = -100;               // pump 3 motor from flush tank
int POWER3 = 100;                // pump 2 motor from sump tank

//Note : We are not using b / constant offset as that will cause problems when flow is zero
//In theory b / constant offset must be zero
const unsigned long HOLD_MS = 5000;   // motor 1 hold time after float opens
const float CAL_A  = 0.02352;         // main flow sensor, L/min per Hz
const float CAL_A2 = 0.02307;         // flush flow sensor, L/min per Hz

const unsigned long GRACE_MS = 500;   // wait after entering mode 2 before checking to avoid immediately shutting off the flush tank while the motor starts up
const float FLUSH_MIN_HZ = 120.0;
// =====================================

//                  L_PWM R_PWM L_EN R_EN
const int M1[4] = { 32,   33,   13,  26 };
const int M2[4] = { 18,   19,    4,  16 };
const int M3[4] = { 23,   25,   21,  22 };

const int FLOW_PIN   = 14;   // main flow sensor (to 10 L tank)
const int FLOW2_PIN  = 27;   // flush tank flow sensor
const int FLOAT_PIN  = 35;   // conducting = LOW
const int BUTTON_PIN = 34;   // button to GND

//counters and timers
volatile unsigned long pulses  = 0;
volatile unsigned long pulses2 = 0;
unsigned long lastPulses = 0;
unsigned long lastPrint = 0;
unsigned long lastPress = 0;
unsigned long lockUntil = 0;
int mode = 0;

// flush-empty check
unsigned long modeStart = 0;
unsigned long chkStart = 0;
unsigned long chkBase = 0;

```

```
//flow sensor interrupt handlers
void IRAM_ATTR countPulse()  { pulses++;  }
void IRAM_ATTR countPulse2() { pulses2++; }

void setupMotor(const int *m) {
  pinMode(m[2], OUTPUT);
  pinMode(m[3], OUTPUT);
  ledcAttach(m[0], 20000, 8);    // driven via LedC peripheral of esp32
  ledcAttach(m[1], 20000, 8);
}

//generic drive command
void drive(const int *m, int percent) {
  int duty = abs(constrain(percent, -100, 100)) * 255 / 100;
  bool on = (percent != 0);
  digitalWrite(m[2], on);
  digitalWrite(m[3], on);
  ledcWrite(m[0], percent < 0 ? duty : 0);   // reverse
  ledcWrite(m[1], percent > 0 ? duty : 0);   // forward
}

void setup() {
  Serial.begin(115200);
  setupMotor(M1);
  setupMotor(M2);
  setupMotor(M3);
  pinMode(FLOAT_PIN, INPUT);
  pinMode(BUTTON_PIN, INPUT_PULLUP);
  pinMode(FLOW_PIN, INPUT_PULLUP);
  pinMode(FLOW2_PIN, INPUT_PULLUP);
  attachInterrupt(digitalPinToInterrupt(FLOW_PIN), countPulse, FALLING);
  attachInterrupt(digitalPinToInterrupt(FLOW2_PIN), countPulse2, FALLING);
}

void loop() {
  // Float: while conducting, keep pushing the lock time forward
  if (digitalRead(FLOAT_PIN) == LOW) lockUntil = millis() + HOLD_MS;
  bool locked = millis() < lockUntil;

  // Button: one step per press (500 ms debounce)
  if (digitalRead(BUTTON_PIN) == LOW && millis() - lastPress > 500) {
    lastPress = millis();
    mode = (mode >= 2) ? 0 : mode + 1;       // 0 -> 1 -> 2 -> 0, and 3 -> 0
    if (mode == 2) { modeStart = millis(); chkStart = 0; }
  }

  // Flush tank empty check (only in mode 2, after the grace period)
  if (mode == 2 && millis() - modeStart >= GRACE_MS) {
    if (chkStart == 0) {                     // start the first 500 ms window
```

```
      chkStart = millis();
      chkBase = pulses2;
    } else if (millis() - chkStart >= 500) {  // window finished
      float rateHz = (pulses2 - chkBase) / 0.5;   // pulses per
    second
      if (rateHz < FLUSH_MIN_HZ) mode = 3;        // tank empty ->
    all off
      chkStart = millis();
      chkBase = pulses2;
    }
  }

  // Motors (mode 1 and mode 3 = all off)
  drive(M1, (mode == 0 && !locked) ? POWER1 : 0);
  drive(M2, (mode == 2) ? POWER2 : 0);
  drive(M3, (mode == 0 || mode == 2) ? POWER3 : 0);

  // Flow readout once per second
  if (millis() - lastPrint >= 1000) {
    lastPrint = millis();
    unsigned long total = pulses;
    unsigned long hz = total - lastPulses;       // pulses in 1 s =
     Hz
    lastPulses = total;

    static unsigned long lastPulses2 = 0;
    unsigned long total2 = pulses2;
    unsigned long hz2 = total2 - lastPulses2;
    lastPulses2 = total2;

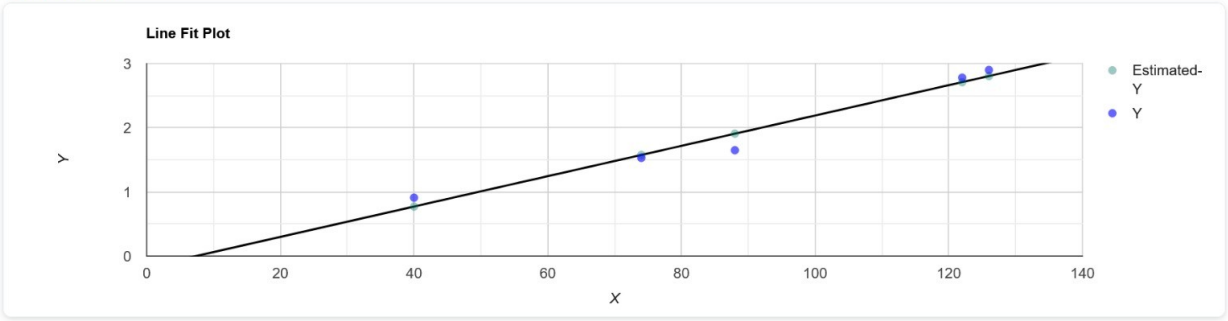
    Serial.print("Flow_mL_s:");    Serial.print(CAL_A * hz * 1000.0
     / 60.0, 2);
    Serial.print(",Total_L:");     Serial.print(total * CAL_A /
    60.0, 3);
    Serial.print(",Flush_mL_s:");  Serial.print(CAL_A2 * hz2 *
    1000.0 / 60.0, 2);
    Serial.print(",Flush_hz:");    Serial.print(hz2);
    Serial.print(",Flush_Total_L:"); Serial.print(total2 * CAL_A2 /
     60.0, 3);
    Serial.print(",Mode:");        Serial.print(mode);
    Serial.print(",Locked:");      Serial.println(locked);
  }
}
```

APA Style

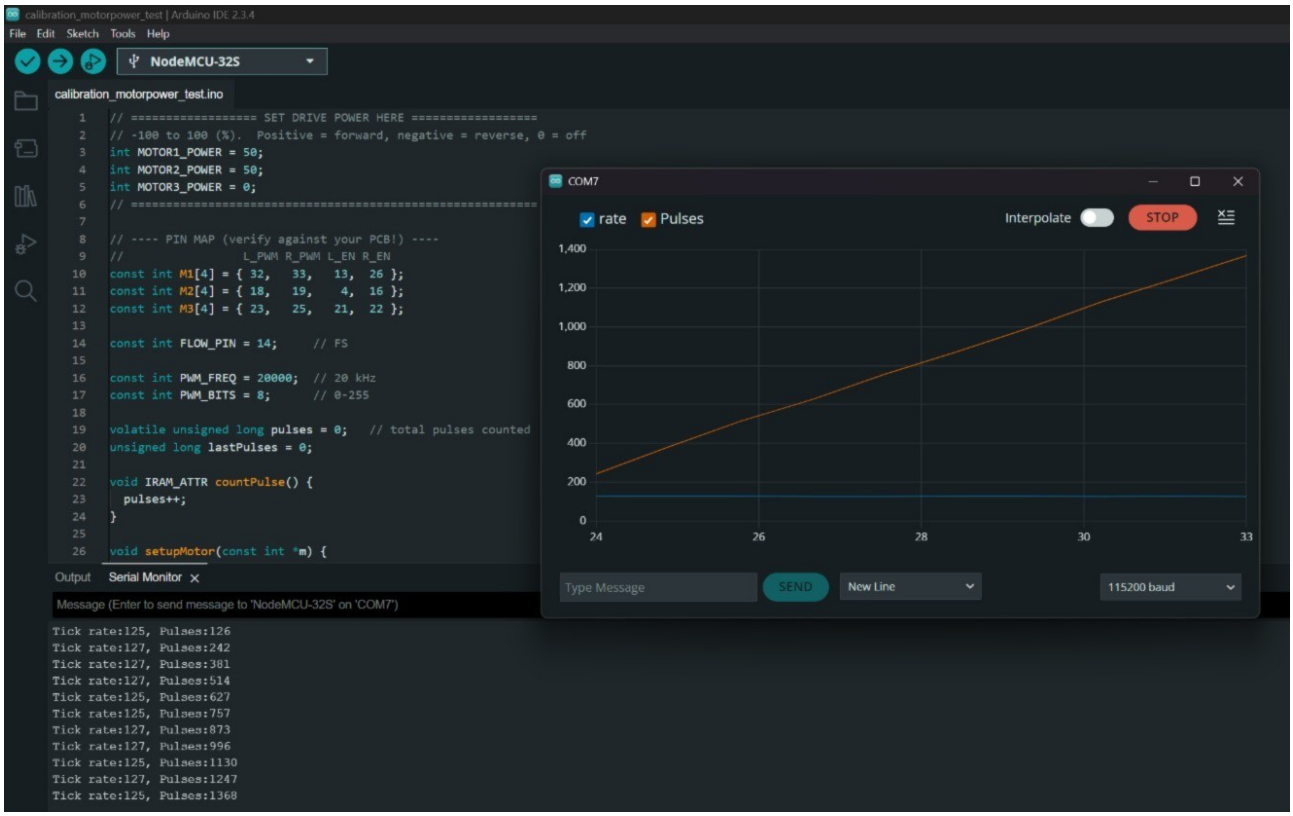
Reporting Linear Regression: Unstandardized (b, Intercept a) and Standardized (β) Coefficients

X predicted Y, R^2 = .97, $F(1,3)$ = 84.4, p = .003, N = 5.
b = 0.024, SE = 0.0026, β = 9.19, 95% CI [0.015, 0.032], p = .003
a = -0.17, SE = 0.25, 95% CI [-0.96, 0.61], p = .531.

Line Fit Plot



Flow sensor tick rate vs Experimental Flow (L/min)



Setup used for calibration. Output from flow sensor and pulse count change is recorded